

EMMANUEL OMONT

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EDUCATION

ETH Zürich <i>Master, DataScience; current GPA: 5.5/6</i> Relevant Coursework: Statistical Modelling, Non Life Insurance, Advanced Algorithms, Mathematical foundation for Data Science, Time Series Analysis, Probability for Artificial Intelligence	Zürich, Switzerland Sep. 2026 – 2028
École Polytechnique Fédérale de Lausanne (EPFL) <i>Bachelor of Science in Computer Science; GPA: 5.82/6</i> Relevant Coursework: Algorithms, Computer Architecture, Machine Learning, Probability, Statistics, Analysis, Algebra	Lausanne, Switzerland Sep. 2023 – 2026
ETH Zürich <i>Exchange Year, Computer Science; current GPA: 5.6/6</i> Relevant Coursework: Big Data, Network Security, Cloud Computing Architecture, Large Language Models	Zürich, Switzerland Sep. 2025 – 2026

EXPERIENCE

Processor Architecture Laboratory (LAP), EPFL <i>Summer in the Lab Research Intern</i> - Contributed to Dynamatic, an open-source dynamically scheduled high-level synthesis (HLS) compiler, over an 8-week research internship - Helped implement a memory optimization pass aimed at reducing circuit latency and area - Adapted the compiler's two buffer placement algorithms to support Control-Free Dataflow Circuits (CFDFCs) produced by the new pass	Lausanne, Switzerland Summer 2025
BoundaryAI <i>Lead Backend Developer</i> - Led backend development for an early-stage startup building AI-powered feedback and survey analysis - Designed and maintained the API powering the company website and the survey analysis pipeline	Lausanne, Switzerland Mar. 2025 – Mar. 2026
EPFL & ETH Zürich <i>Teaching Assistant</i> - ETH: Teaching Assistant for Systems Programming and Computer Architecture (2nd-year B.Sc.) and Big Data for Engineers (M.Sc.) - EPFL: Student Assistant for Computer Science for Mathematicians (2024–25), Mechanics (Fall 2024) and Discrete Mathematics (Spring 2025) - Led exercise sessions, re-explained lecture material and answered students' questions	Lausanne & Zürich, Switzerland Sep. 2024 – Present
Machine Learning and Optimization Laboratory, EPFL <i>Project Assistant</i> - Developed frontend features for Disco (discolab.ai), a decentralized, privacy-preserving collaborative ML platform built in collaboration with Yale University - Improved software reliability by resolving GitHub issues and added tooling for testing trained models	Lausanne, Switzerland 2024 – 2025
Investment Society UNIL–EPFL <i>Financial Analyst</i> Produced equity research and proposed investment strategies focused on the networking and technology sectors	Lausanne, Switzerland Mar. 2025 – Aug. 2025

PROJECTS

CrisisCommand <i>Web App, Infrastructure</i> - Built a MUN crisis management software, that scaled to 300+ users, and 10+ paid conferences 2 weeks after launch - Recieved a buy offer from the CEO of MyMUN (300k+ users), 1 month after launch.	2025
Clastrix <i>Web App, Open Source</i> Built an open-source Kahoot-style platform for interactive quizzes with separate presenter and participant screens and Markdown support; handled 100+ concurrent users	2025
Homelab <i>Infrastructure, Self-hosting</i> Built and administer a personal server cluster of 10+ machines with 7.5+ TB of storage and 200+ GB of RAM, hosting all personal projects	2025 – Present
DatIN <i>Data Collection & Analysis</i> Built a data pipeline collecting records on 2M+ Swiss companies and ran exploratory analysis on the resulting dataset	2025
MemeHater <i>Machine Learning</i> Fine-tuned a VisualBERT multimodal model with PyTorch and TensorFlow to detect hateful memes	2024

LEADERSHIP & ACTIVITIES

Zürich MUN – USG of Chairs & Academics: Recruited and managed a team of 15 to supervise the debates	2025 – 2026
MUN EPFL – IT Manager: run the association's website and servers	2024 – 2026
S4S, EPFL – Student Assistant: taught intro sessions to incoming students	Sep. 2024

SKILLS & LANGUAGES

Languages: Python, R, C/C++, Java, Scala, JavaScript/TypeScript (Node.js), Assembly, Rust
Frameworks: Vue/Nuxt, React/Next, Svelte, PyTorch, TensorFlow
Areas: Systems Programming, Machine Learning, Backend & APIs, Data Analysis, Networking & Cybersecurity
Spoken: French (native), German (C1, certified by Goethe-Institut), English (fluent), Spanish (fluent)